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CUET UG

Chemistry

Solutions

Official Syllabus Unit I · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET UG Chemistry, Unit I)	Skills the questions test
Types of Solutions & Concentration	Types of solutions, expression of concentration of solutions of solids in liquids.	Interconverting Molarity, Molality, Mole Fraction, and Mass %; identifying temperature-independent units.
Solubility & Henry's Law	Solubility of gases in liquids, solid solutions, Henry's law.	Applying $p = K_H x$; analyzing temperature scaling of K_H; explaining scuba bends, soda packaging, and anoxia.
Raoult's Law & Deviations	Raoult's law, ideal and non-ideal solutions, positive and negative deviations, azeotropes.	Calculating partial and total vapour pressures; identifying intermolecular forces (f_{AB} vs f_{AA}); classifying azeotropes.
Colligative Properties	Relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, osmotic pressure, reverse osmosis.	Applying colligative equations to compute solute molar masses (M_2); evaluating isotonicity and membrane transport.
Abnormal Masses & Van 't Hoff	Abnormal molecular mass, van 't Hoff factor (i), degree of association and dissociation.	Calculating i from α or β; predicting ranking of colligative effects for electrolyte mixtures.

EXAM PATTERN

CUET UG Exam Pattern & Strategy: The NTA CUET UG Chemistry paper contains **50 compulsory questions** to be solved in **60 minutes** (+5 for correct, −1 for incorrect). Physical chemistry opens with Unit I Solutions, delivering 4–5 numerical and conceptual questions focusing on concentration conversions, Henry's law, azeotropic deviations, and colligative equations corrected with the van 't Hoff factor i.

Concept Module 1: Types of Solutions & Concentration Metrics

A **solution** is a homogeneous mixture of two or more chemically non-reacting substances whose composition can be varied within certain limits.

- **Solvent:** The component present in the largest quantitative proportion, dictating the physical state of the solution.
- **Solute:** The substance dissolved in the solvent (one or more components).
- **Binary Solution:** A solution consisting of exactly two components (one solute and one solvent).

The Nine Types of Binary Solutions

Solute	Solvent	Solution Type	Standard NCERT Benchmark Examples
Gas	Gas	Gaseous Solution	Air (mixture of O_2 and N_2 gases)
Liquid	Gas	Gaseous Solution	Chloroform vapour mixed with nitrogen gas; humidity
Solid	Gas	Gaseous Solution	Camphor vapours in nitrogen gas; smoke
Gas	Liquid	Liquid Solution	Oxygen dissolved in water; carbonated soda drinks
Liquid	Liquid	Liquid Solution	Ethanol dissolved in water; vinegar
Solid	Liquid	Liquid Solution	Glucose, urea, or NaCl dissolved in water
Gas	Solid	Solid Solution	Hydrogen gas adsorbed on Palladium surface ($\frac{H_2}{P}d$)
Liquid	Solid	Solid Solution	Amalgam of mercury with sodium ($Hg - Na$)
Solid	Solid	Solid Solution	Copper dissolved in gold; Brass (Copper + Zinc alloy)

Quantitative Concentration Formulations

Concentration defines the quantity of solute dissolved in a specific amount of solvent or solution:

1. **Mass Percentage ($\%\frac{w}{w}$):** Mass of solute per 100 g of solution:

$$\%\left(\frac{w}{w}\right) = \frac{\text{Mass of solute } (w_2)}{\text{Total mass of solution } (w_1 + w_2)} \times 100$$

2. **Volume Percentage ($\%\frac{v}{v}$):** Volume of solute per 100 mL of solution:

$$\%\left(\frac{v}{v}\right) = \frac{\text{Volume of solute}}{\text{Total volume of solution}} \times 100$$

A $35\%\left(\frac{v}{v}\right)$ solution of ethylene glycol in water is employed as an automotive coolant/antifreeze, depressing the freezing point of water to 255.4 K ($-17.6°C$).

3. **Mass by Volume Percentage ($\%\frac{w}{v}$):** Mass of solute in grams per 100 mL of solution:

$$\%\left(\frac{w}{v}\right) = \frac{\text{Mass of solute in g}}{\text{Volume of solution in mL}} \times 100$$

4. **Parts Per Million (ppm):** Employed for trace solute quantities (e.g. atmospheric pollution, water hardness):

$$\text{ppm} = \frac{\text{Number of parts of solute}}{\text{Total number of parts of all components}} \times 10^6 = \frac{w_2}{w_{\text{solution}}} \times 10^6$$

5. **Mole Fraction (x):** Ratio of moles of a given component to total moles present:

$$x_1 = \frac{n_1}{n_1 + n_2}, \quad x_2 = \frac{n_2}{n_1 + n_2}$$

For any solution containing k components: $\sum_{i=1}^{k} x_i = 1$. It is a dimensionless, temperature-independent quantity.

6. **Molarity (M):** Number of moles of solute dissolved per litre (1 dm^3) of solution:

$$M = \frac{\text{Moles of solute } (n_2)}{\text{Volume of solution in Litres } (V_L)} = \frac{w_2 \times 1000}{M_2 \times V_{\text{mL}}} \quad [\text{units: mol/L} \vee \text{M}]$$

7. **Molality (m):** Number of moles of solute dissolved per kilogram (1000 g) of solvent:

$$m = \frac{\text{Moles of solute } (n_2)}{\text{Mass of solvent in kg } (W_1)} = \frac{w_2 \times 1000}{M_2 \times w_1(\text{in g})} \quad [\text{units: mol/kg} \vee \text{m}]$$

INTERCONVERSION: MOLALITY (M) AND MOLARITY (M)

Given solution density d in g/mL (or g/cm^3), molarity M, and solute molar mass M_2:

$$m = \frac{1000 \times M}{1000 \times d - M \times M_2}$$

Conversely:

$$M = \frac{1000 \times d \times m}{1000 + m \times M_2}$$

CRUCIAL CUET DISTINCTION: TEMPERATURE DEPENDENCE

- **Temperature-Independent Units: Molality (m), Mole Fraction (x), Mass Percentage ($\%\frac{w}{w}$)**, and **ppm** depend solely on mass. Since mass is invariant with temperature, these metrics remain constant when temperature changes.
- **Temperature-Dependent Units: Molarity (M), Normality (N)**, and **Volume Percentage ($\%\frac{v}{v}$)** depend directly on the volume of the solution. Because liquids expand upon heating, **Molarity decreases as temperature increases!**

CONCENTRATION UNIT INTERCONVERSION

Problem: An aqueous solution of glucose ($C_6H_{12}O_6$, $M_2 = 180$ g/mol) is labeled as $10\%(\frac{w}{w})$. The density of the solution is 1.20 g/mL. Calculate: (a) Molality (m), (b) Mole fraction of glucose (x_2), (c) Molarity (M).

Solution: Take 100 g of solution: mass of glucose $w_2 = 10$ g, mass of water $w_1 = 90$ g $= 0.090$ kg. Moles of glucose $n_2 = \frac{10}{180} = 0.0556$ mol. Moles of water $n_1 = \frac{90}{18} = 5.00$ mol.

(a) **Molality:** $m = \frac{n_2}{W_1}(\text{kg}) = \frac{0.0556}{0.090} = 0.617$ m.

(b) **Mole Fraction:** $x_2 = \frac{n_2}{n_1+n_2} = \frac{0.0556}{5.00+0.0556} = \frac{0.0556}{5.0556} = 0.0110$.

(c) **Molarity:** Volume of 100 g solution $V = \frac{\text{mass}}{\text{density}} = \frac{100}{1.20} = 83.33$ mL $= 0.08333$ L. $M = \frac{n_2}{V_L} = \frac{0.0556}{0.08333} = 0.667$ M.

Concept Module 2: Solubility & Henry's Law

Solubility is the maximum amount of solute that can dissolve in a specified amount of solvent at a given temperature and pressure.

- **Saturated Solution:** A solution in which no more solute can dissolve at that specific temperature. A dynamic equilibrium exists:

$$\text{Solute} + \text{Solvent} \leftrightarrow \text{Solution}$$

- **Unsaturated Solution:** A solution containing less than the saturation threshold of solute.

Solubility of a Solid in a Liquid

- **Nature of Solute and Solvent:** Governed by the fundamental thermodynamic rule: **"Like dissolves like"**. Polar solutes (ionic salts like $NaCl, KNO_3$) dissolve readily in polar solvents (H_2O), while non-polar solutes (I_2, naphthalene, anthracene) dissolve in non-polar organic solvents (benzene, CCl_4).
- **Effect of Temperature:**
  - If dissolution is **endothermic** ($\Delta_{sol}H > 0$, e.g. $KNO_3, NaCl, KCl$): solubility **increases** with rising temperature according to Le Chatelier's principle.
  - If dissolution is **exothermic** ($\Delta_{sol}H < 0$, e.g. $Ce_2(SO_4)_3$, lithium salts): solubility **decreases** with rising temperature.
- **Effect of Pressure:** Pressure exerts **no noticeable effect** on the solubility of solids in liquids because solids and liquids are practically incompressible.

Solubility of a Gas in a Liquid: Henry's Law

The solubility of any gas in a liquid increases with increase in pressure. William Henry quantitatively formulated this relationship:

HENRY'S LAW FORMULATIONS

"The partial pressure of the gas in vapour phase (p) is directly proportional to the mole fraction of the gas (x) in the solution."

$$p = K_H \times x$$

where K_H is the **Henry's Law Constant**.

- A plot of partial pressure (p) against mole fraction (x) yields a straight line passing through the origin with slope equal to K_H.

HIGH-YIELD HENRY'S LAW PRINCIPLES & EXAM TRAPS

1. **Inverse Solubility Scaling:** At a given pressure p, higher K_H implies **lower solubility** of the gas in liquid:
$$x = \frac{p}{K_H}$$
2. **Temperature Scaling of K_H:** For a given gas, K_H **increases** as temperature rises. Consequently, the solubility of gases in liquids **decreases** with increasing temperature.
3. **Aquatic Life Adaptation:** Aquatic species (fish, amphibians) thrive far better in cold waters than in warm waters because dissolved oxygen concentration is significantly higher at lower temperatures.
4. **Order of K_H values at 293 K (Solubility Order):**
$$K_H : He(144.97) > H_2(69.16) > N_2(76.48) > O_2(34.86) > Ar(40.3 \text{ kbar})$$

Therefore, oxygen (O_2) is over twice as soluble in water as nitrogen (N_2).

Practical Applications of Henry's Law

1. **Carbonated Soda Beverages:** Soft drink bottles are sealed under high pressure to maximize the solubility of CO_2 gas. When opened, pressure drops to 1 atm, reducing solubility and producing rapid gas effervescence.
2. **Deep-Sea Diving ("The Bends"):** As deep-sea scuba divers descend, high hydrostatic pressure forces atmospheric nitrogen (N_2) to dissolve heavily into blood and tissues. When ascending rapidly, pressure plummets, causing dissolved N_2 to release abruptly as bubbles in capillaries. These bubbles block circulation, producing painful and fatal decompression sickness known as **the bends**.

- **Medical Countermeasure:** Diver breathing tanks are diluted with **Helium** ($11.7\%He, 56.2\%N_2, 32.1\%O_2$) because Helium has an extremely high K_H and very low solubility in blood.

3. **High-Altitude Anoxia:** At high altitudes, atmospheric oxygen partial pressure is lower than at sea level. This leads to low oxygen concentrations in the blood and brains of climbers, impairing cognitive function and producing a clinical condition called **anoxia**.

Concept Module 3: Raoult's Law & Binary Vapour Pressures

Vapour pressure is the equilibrium pressure exerted by the vapour phase above a liquid surface at a constant temperature.

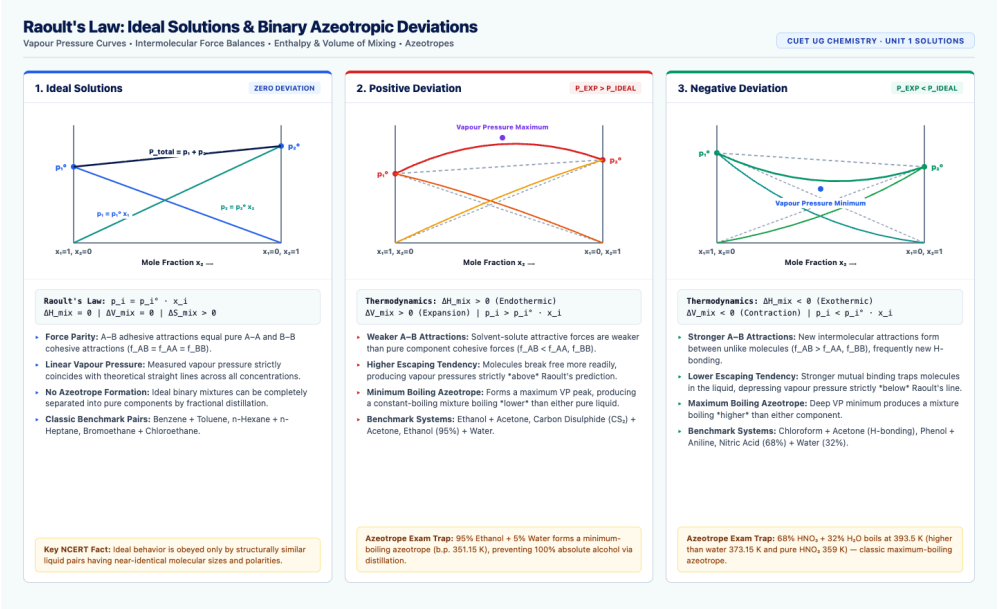


Figure 1: Raoult's Law on Vapour Pressure: Linear ideal behaviour vs positive deviation (minimum-boiling azeotrope) and negative deviation (maximum-boiling azeotrope).

Raoult's Law for Liquid-Liquid Volatile Mixtures

For a binary solution of two volatile miscible liquids 1 and 2: **"For any volatile component in a solution, its partial vapour pressure is directly proportional to its mole fraction in the solution."**

$$p_1 = p_1^0 \times x_1, \quad p_2 = p_2^0 \times x_2$$

where p_1^0 and p_2^0 represent the vapour pressures of pure components 1 and 2 at the same temperature.

Applying **Dalton's Law of Partial Pressures**, the total vapour pressure P_{total} is:

$$P_{\text{total}} = p_1 + p_2 = p_1^0 x_1 + p_2^0 x_2 = p_1^0(1 - x_2) + p_2^0 x_2$$

$$P_{\text{total}} = p_1^0 + (p_2^0 - p_1^0) \times x_2$$

Composition of Vapour Phase (Dalton's Gas Formulation)

If y_1 and y_2 denote the mole fractions of components 1 and 2 in the **vapour phase**:

$$p_1 = y_1 \times P_{\text{total}} \implies y_1 = \frac{p_1}{P_{\text{total}}} = \frac{p_1^0 x_1}{P_{\text{total}}}$$

$$p_2 = y_2 \times P_{\text{total}} \implies y_2 = \frac{p_2}{P_{\text{total}}} = \frac{p_2^0 x_2}{P_{\text{total}}}$$

Konovalov's Rule: The vapour phase is always richer in the component that is more volatile (possessing higher pure vapour pressure p^0).

Concept Module 4: Ideal and Non-Ideal Binary Solutions

Binary liquid mixtures are categorized according to their adherence to Raoult's law over the entire range of concentration:

1. Ideal Solutions

An ideal solution satisfies Raoult's law at all concentrations and temperatures:

- **Thermodynamic Criteria:**
  1. $p_1 = p_1^0 x_1$ and $p_2 = p_2^0 x_2$ (Obeys Raoult's law).
  2. $\Delta_{\text{mix}} H = 0$ (Zero heat absorbed or evolved upon mixing).
  3. $\Delta_{\text{mix}} V = 0$ (Total volume strictly equals the sum of component volumes).
  4. $\Delta_{\text{mix}} S > 0$ and $\Delta_{\text{mix}} G < 0$ (Spontaneous mixing).
- **Molecular Basis:** Intermolecular attractive forces between unlike molecules ($A - B$) are identical in strength to those between like molecules ($A - A$ and $B - B$):

$$f_{A-B} = f_{A-A} = f_{B-B}$$

- **NCERT Benchmark Examples:**
  - Benzene + Toluene
  - *n*-Hexane + *n*-Heptane
  - Bromoethane + Chloroethane
  - Chlorobenzene + Bromobenzene

2. Non-Ideal Solutions: Positive Deviation

- **Observed Behaviour:** Measured vapour pressure is strictly **higher** than predicted by Raoult's law ($P_{\text{total}} > p_1^0 x_1 + p_2^0 x_2$).
- **Molecular Basis:** The $A - B$ intermolecular attractive forces are **weaker** than pure $A - A$ and $B - B$ cohesive forces:

$$f_{A-B} < f_{A-A}, \quad f_{A-B} < f_{B-B}$$

Molecules have a higher escaping tendency into the vapour phase.

- **Thermodynamic Criteria:**

$$\Delta_{\text{mix}} H > 0 \quad \text{[Endothermic mixing: absorbs heat, solution cools]}$$

$$\Delta_{\text{mix}} V > 0 \quad \text{[Volume expansion: total volume exceeds sum of components]}$$

- **NCERT Benchmark Pairs:**
  - Ethanol + Acetone (Acetone molecules disrupt strong intermolecular H-bonding between ethanol molecules).
  - Carbon Disulphide (CS_2) + Acetone (Dipole-dipole attractions of acetone are weakened).
  - Ethanol + Water, Acetone + Benzene, Carbon tetrachloride (CCl_4) + Chloroform ($CHCl_3$).

3. Non-Ideal Solutions: Negative Deviation

- **Observed Behaviour:** Measured vapour pressure is strictly **lower** than predicted by Raoult's law ($P_{\text{total}} < p_1^0 x_1 + p_2^0 x_2$).
- **Molecular Basis:** The $A - B$ intermolecular attractive forces are **stronger** than pure $A - A$ and $B - B$ cohesive forces:

$$f_{A-B} > f_{A-A}, \quad f_{A-B} > f_{B-B}$$

Frequently, **new hydrogen bonds** form between unlike components that neither pure liquid possessed.

- **Thermodynamic Criteria:**

$$\Delta_{\text{mix}} H < 0 \quad \text{[Exothermic mixing: releases heat, solution warms]}$$

$$\Delta_{\text{mix}} V < 0 \quad \text{[Volume contraction: total volume is less than sum of components]}$$

- **NCERT Benchmark Pairs:**
  - Chloroform ($CHCl_3$) + Acetone (CH_3COCH_3): A strong new hydrogen bond forms between the acidic proton of chloroform and the carbonyl oxygen of acetone:
$$Cl_3C - H \cdots O = C(CH_3)_2$$
  - Phenol + Aniline: Hydrogen bonding between phenolic $-OH$ proton and aniline $-NH_2$ nitrogen.
  - Nitric acid (HNO_3) + Water (H_2O), Hydrochloric acid (HCl) + Water (H_2O).

Azeotropic Mixtures (Constant Boiling Mixtures)

An **azeotrope** is a binary liquid mixture having identical composition in both liquid and vapour phases and boiling at a constant temperature without fractional distillation separation.

1. **Minimum Boiling Azeotropes:** Formed by mixtures exhibiting **large positive deviations** from Raoult's law. The vapour pressure curve passes through a maximum, meaning the boiling point passes through a minimum.
   - **NCERT Example:** Mixture of 95.4%$\left(\frac{v}{v}\right)$ ethanol and 4.6%$\left(\frac{v}{v}\right)$ water. It boils at 351.15 K, which is lower than pure ethanol (351.5 K) and pure water (373.15 K). Fractional distillation cannot yield absolute (100%) alcohol.
2. **Maximum Boiling Azeotropes:** Formed by mixtures exhibiting **large negative deviations** from Raoult's law. The vapour pressure curve passes through a minimum, meaning the boiling point reaches a maximum.
   - **NCERT Example:** Mixture of 68%$\left(\frac{w}{w}\right)HNO_3$ and 32%$\left(\frac{w}{w}\right)H_2O$. It boils at 393.5 K, which is higher than pure nitric acid (359 K) and pure water (373.15 K).

Concept Module 5: The Four Colligative Properties

Colligative properties are physical properties of dilute solutions that depend solely on the **number of solute particles** (molecules or ions) present in solution and are completely **independent of their chemical nature or identity**.

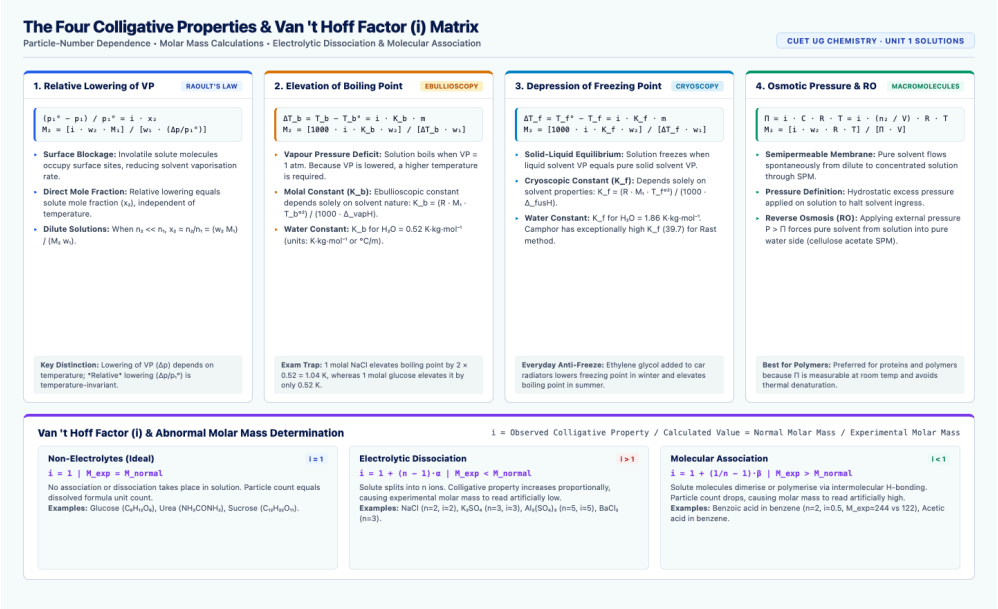


Figure 2: The Four Colligative Properties & Van 't Hoff Factor Matrix: Mathematical equations, molar mass determinations, and dissociation/association equilibria.

1. Relative Lowering of Vapour Pressure (RLVP)

When an involatile, non-electrolyte solute is dissolved in a volatile solvent, solute particles block part of the surface, diminishing the escaping rate of solvent molecules.

- Vapour pressure of solution $p_1 = p_1^0 x_1 = p_1^0(1 - x_2)$.

- Lowering of vapour pressure: $\Delta p = p_1^0 - p_1 = p_1^0 x_2$.
- **Relative Lowering of Vapour Pressure:**

$$\frac{\Delta p}{p_1^0} = \frac{p_1^0 - p_1}{p_1^0} = x_2 = \frac{n_2}{n_1 + n_2}$$

For dilute solutions where $n_2 \ll n_1$:

$$\frac{p_1^0 - p_1}{p_1^0} \approx \frac{n_2}{n_1} = \frac{\frac{w_2}{M_2}}{\frac{w_1}{M_1}} = \frac{w_2 \times M_1}{M_2 \times w_1}$$

$$M_2 = \frac{w_2 \times M_1}{w_1 \times \left(\frac{p_1^0 - p_1}{p_1^0}\right)}$$

2. Elevation of Boiling Point (Ebullioscopy)

The boiling point of a liquid is the temperature at which its vapour pressure becomes equal to the external atmospheric pressure (1.013 bar = 1 atm). Because an involatile solute depresses vapour pressure, the solution must be heated to a higher temperature to reach atmospheric pressure.

- Boiling point elevation: $\Delta T_b = T_b - T_b^0$.
- For dilute solutions, ΔT_b is directly proportional to molal concentration (m):

$$\Delta T_b = K_b \times m = \frac{1000 \times K_b \times w_2}{M_2 \times w_1}$$

where K_b is the **Molal Elevation Constant** or **Ebullioscopic Constant**.

- Units of K_b: K kg mol^{-1} or °C kg mol^{-1} or °C/m.
- Thermodynamic formula for K_b:

$$K_b = \frac{R \times M_1 \times \left(T_b^0\right)^2}{1000 \times \Delta_{\text{vap}} H}$$

For water (H_2O): $K_b = 0.52$ K kg mol^{-1}.

3. Depression of Freezing Point (Cryoscopy)

The freezing point of a substance is the temperature at which the liquid phase and solid phase exert identical vapour pressures. Since solute lowers the solvent's vapour pressure, the solution's liquid curve intersects the solid solvent curve at a lower temperature.

- Freezing point depression: $\Delta T_f = T_f^0 - T_f$.

$$\Delta T_f = K_f \times m = \frac{1000 \times K_f \times w_2}{M_2 \times w_1}$$

where K_f is the **Molal Depression Constant** or **Cryoscopic Constant**.

- Units of K_f: K kg mol^{-1}.
- Thermodynamic formula for K_f:

$$K_f = \frac{R \times M_1 \times \left(T_f^0\right)^2}{1000 \times \Delta_{\text{fus}} H}$$

For water (H_2O): $K_f = 1.86$ K kg mol^{-1}.

4. Osmotic Pressure (Π) & Reverse Osmosis

- **Osmosis:** The spontaneous net movement of solvent molecules from a region of lower solute concentration (pure solvent or dilute solution) to a region of higher solute concentration through a **semi-permeable membrane (SPM)**.
- **Semi-Permeable Membrane (SPM):** A continuous membrane containing sub-microscopic pores that allow small solvent molecules (H_2O) to pass while blocking larger solute particles/hydrated ions (e.g. pig's bladder, parchment, synthetic **cellulose acetate**).

- **Osmotic Pressure (Π):** The exact excess hydrostatic pressure that must be applied to the solution side to prevent the inward flux of solvent through the SPM.
- **Van 't Hoff Equation for Dilute Solutions:**

$$\Pi = CRT = \left(\frac{n_2}{V}\right)RT = \frac{w_2RT}{M_2V}$$

$$M_2 = \frac{w_2RT}{\Pi V}$$

where C is molar concentration (mol/L), $R = 0.0821$ L atm $K^{-1}mol^{-1} = 8.314$ J $K^{-1}mol^{-1}$, and T is temperature in Kelvin.

Types of Solutions Based on Osmotic Pressure

1. **Isotonic Solutions:** Two solutions having identical osmotic pressures at the same temperature ($\Pi_1 = \Pi_2 \implies C_1 = C_2$). No net osmosis occurs across an SPM.
   - **Intravenous Injection Benchmark:** Normal saline solution is $0.9\%(\frac{w}{v})NaCl$ in water. It is strictly **isotonic** with human blood plasma.
2. **Hypertonic Solution:** A solution having **higher** osmotic pressure than intracellular fluid. If red blood cells (RBCs) are placed in $> 0.9\%NaCl$ solution, water flows **out** of cells, causing them to shrink and shrivel (**crenation**).
3. **Hypotonic Solution:** A solution having **lower** osmotic pressure than intracellular fluid. If RBCs are placed in $< 0.9\%NaCl$ (or pure distilled water), water rushes **into** cells, causing them to swell and burst (**haemolysis**).

Reverse Osmosis (RO) and Water Desalination

If an external mechanical pressure **greater than the osmotic pressure** ($P > \Pi$) is applied to the concentrated solution compartment, the natural osmotic flow is reversed! Solvent molecules are forced from the solution (e.g. seawater) through the SPM into the pure water reservoir.

- **Membrane Material:** Modern RO desalination plants utilize porous films of **Cellulose Acetate** supported on an inert porous polyacrylonitrile matrix.

WHY OSMOTIC PRESSURE IS SUPERIOR FOR MACROMOLECULES

Osmotic pressure is overwhelmingly preferred over ΔT_b or ΔT_f for determining the molar masses of polymers, proteins, DNA, and biomolecules:

1. Osmotic pressure measurements are executed at **ambient room temperature** (298 K), preventing thermal denaturation of sensitive biomolecules.
2. Molar concentration (M) is utilized instead of molality (m).
3. The magnitude of Π is very large and readily measurable (several millimeters of liquid column) even for extremely dilute polymer solutions where ΔT_b and ΔT_f are too miniscule ($\approx 10^{-3}$ K) to be measured accurately.

Concept Module 6: Abnormal Molar Masses & Van 't Hoff Factor (i)

When a solute undergoes electrolytic dissociation or molecular association in solution, the actual number of particles differs from the number of dissolved formula units. Because colligative properties depend solely on total particle count, the experimentally measured colligative property deviates from the theoretical value, yielding an **abnormal molar mass**.

Jacobus Henricus van 't Hoff introduced the **Van 't Hoff Factor (i)** to account for these deviations:

DEFINITIONS OF VAN 'T HOFF FACTOR (I)

$$i = \frac{\text{Experimental (Observed) Colligative Property}}{\text{Calculated (Theoretical) Colligative Property}}$$

$$i = \frac{\text{Actual number of particles after association/dissociation}}{\text{Number of formula units initially dissolved}}$$

Since colligative property is inversely proportional to solute molar mass (C.P. $\propto \frac{1}{M_2}$):

$$i = \frac{\text{Normal (Theoretical) Molar Mass}}{\text{Abnormal (Experimental) Molar Mass}} = \frac{M_{\text{normal}}}{M_{\text{obs}}}$$

All Modified Colligative Equations

1. Relative Lowering of Vapour Pressure: $\frac{p_1^0 - p_1}{p_1^0} = i \times x_2$
2. Elevation of Boiling Point: $\Delta T_b = i \times K_b \times m$
3. Depression of Freezing Point: $\Delta T_f = i \times K_f \times m$
4. Osmotic Pressure: $\Pi = i \times C \times R \times T$

Degree of Dissociation (α)

Consider an electrolyte A_xB_y dissociating into n ions ($n = x + y$):

$$A_xB_y \leftrightarrow xA^{y+} + yB^{x-}$$

- Initial moles at $t = 0$: 1 mole of A_xB_y, 0 ions.
- Equilibrium moles: $(1 - \alpha)$ moles of undissociated solute, $n\alpha$ moles of ions.
- Total particles at equilibrium: $(1 - \alpha) + n\alpha = 1 + (n - 1)\alpha$.

$$i = \frac{1 + (n-1)\alpha}{1} \Longrightarrow \alpha = \frac{i - 1}{n - 1}$$

- For non-electrolytes (glucose, urea): $i = 1, \alpha = 0$.
- For complete dissociation ($\alpha = 1$): $i = n$.
  - $NaCl \rightarrow Na^+ + Cl^- \Longrightarrow n = 2, i = 2$.
  - $K_2SO_4 \rightarrow 2K^+ + SO_4^{2-} \Longrightarrow n = 3, i = 3$.
  - $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-} \Longrightarrow n = 5, i = 5$.
  - $K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-} \Longrightarrow n = 5, i = 5$.

Degree of Association (β)

Consider n molecules of solute associating to form a polymer $(A)_n$:

$$nA \leftrightarrow A_n$$

- Initial moles: 1 mole of A, 0 associated polymer.
- Equilibrium moles: $(1 - \beta)$ moles of monomer, $\frac{\beta}{n}$ moles of polymer.
- Total particles: $(1 - \beta) + \frac{\beta}{n} = 1 - \left(1 - \frac{1}{n}\right)\beta$.

$$i = 1 - \left(1 - \frac{1}{n}\right)\beta \Longrightarrow \beta = \frac{1 - i}{1 - \frac{1}{n}}$$

- For complete dimerization ($n = 2, \beta = 1$): $i = 1 - \frac{1}{2} = 0.5$.
- **Classic Dimerization Example:** Benzoic acid (C_6H_5COOH) or Acetic acid (CH_3COOH) dissolved in **benzene**. Carboxylic acids dimerise via intermolecular hydrogen bonding:

$$2C_6H_5COOH \leftrightarrow (C_6H_5COOH)_2$$

Because particle count is halved, the observed molar mass is approximately double the normal molar mass:

$$M_{\text{normal}} = 122 \text{ g/mol}, \quad M_{\text{obs}} \approx 244 \text{ g/mol}, \quad i \approx 0.5$$

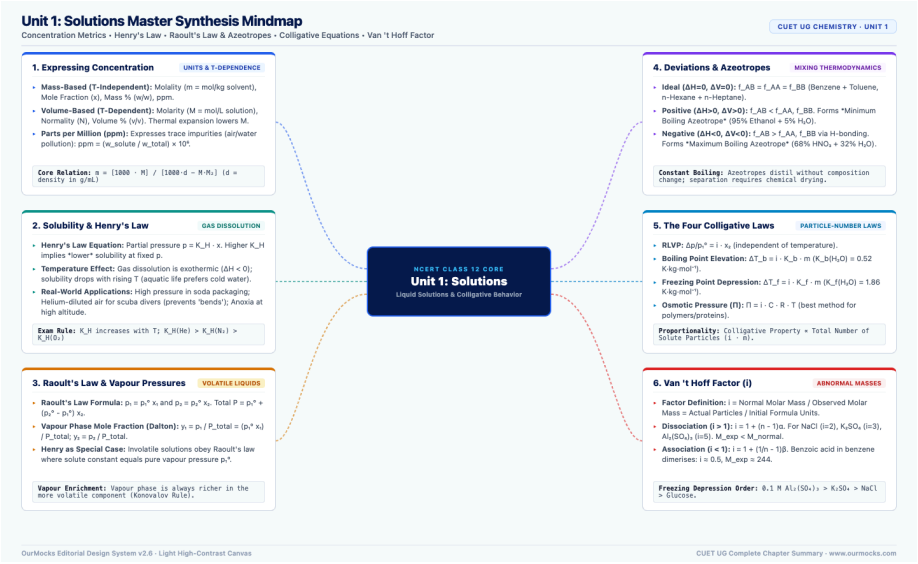


Figure 3: Master Synthesis Architecture: Complete Unit I cognitive roadmap integrating solution classifications, Raoult's law deviations, colligative mechanics, and van 't Hoff factor calculations.

MASTER SYNTHESIS MAP & PROBLEM-SOLVING DIAGNOSTICS

- **Raoult's Law as a Limiting Case of Henry's Law:** In any real binary solution, as the solute mole fraction approaches zero ($x_2 \to 0$), the solute obeys Henry's law ($p_2 = K_H x_2$), while the solvent ($x_1 \to 1$) simultaneously obeys Raoult's law ($p_1 = p_1^0 x_1$). Raoult's law is mathematically identical to Henry's law with $K_H = p_1^0$.
- **Azeotropic Phase Boundaries:** Azeotropes boil at constant temperature and distill without any composition change ($y_i = x_i$). Minimum-boiling azeotropes stem from large positive deviations (ethanol-water 95.4%), whereas maximum-boiling azeotropes stem from strong negative deviations ($68\% HNO_3 + 32\% H_2O$).
- **Colligative Equivalence & Ranking Rules:** For equimolar (0.1 M) aqueous solutions, colligative magnitudes ($\Delta T_b, \Delta T_f, \Pi$, RLVP) follow the exact order of total particle concentration: $i \times C$. Thus: $Al_2(SO_4)_3 (i=5) > K_2SO_4 (i=3) = BaCl_2 (i=3) > NaCl (i=2) > \frac{\text{Urea}}{\text{Glucose}} (i=1)$.
- **Abnormal Molecular Mass Diagnostics:** Solute association (e.g. benzoic acid in benzene) reduces total particles ($i < 1$), doubling apparent molar mass, whereas ionic dissociation (e.g. KCl in water) multiplies particles ($i > 1$), halving apparent molar mass.

UNIT I MASTER FORMULA & HIGH-YIELD CHEATSHEET

Core Law / Property	Mathematical Expression	CUET Problem Strategy & Constraints
Concentration Conversion	$m = \frac{1000M}{1000d - MM_2}$	Molality (m), mole fraction (x), and mass % are invariant with temperature; Molarity decreases with rising T.
Henry's Law	$p = K_H x$ (K_H in bar / atm)	Higher $T \rightarrow$ higher $K_H \rightarrow$ lower gas solubility. Scuba mix: $11.7\% He + 56.2\% N_2 + 32.1\% O_2$.
Raoult's Law (Volatile)	$p_1 = p_1^0 x_1, \quad p_2 = p_2^0 x_2$	Total pressure: $P_{\text{total}} = p_1^0 + (p_2^0 - p_1^0)x_2$. Vapour mole fraction: $y_1 = \frac{p_1}{P_{\text{total}}}$.
Positive Deviation	$f_{AB} < f_{AA}, f_{BB}; \Delta H > 0, \Delta V > 0$	Minimum-boiling azeotrope (e.g. 95.4% ethanol +4.6% water, b.p. 351.15 K, acetone + CS_2).
Negative Deviation	$f_{AB} > f_{AA}, f_{BB}; \Delta H < 0, \Delta V < 0$	Maximum-boiling azeotrope (e.g. $68\% HNO_3 + 32\% H_2O$, b.p. 393.5 K, chloroform + acetone).
RLVP (Colligative)	$\frac{p_1^0 - p_1}{p_1^0} = ix_2 \approx i\left(\frac{n_2}{n_1}\right)$	Relative lowering equals mole fraction of solute; independent of solvent identity when expressed as x_2.
Boiling Elevation	$\Delta T_b = iK_b m; \quad M_2 = \frac{1000 K_b w_2}{\Delta T_b w_1}$	K_b (ebullioscopic constant) depends solely on solvent: $K_{b(H_2O)} = 0.52$ K kg mol^{-1}.
Freezing Depression	$\Delta T_f = iK_f m; \quad M_2 = \frac{1000 K_f w_2}{\Delta T_f w_1}$	Antifreeze: ethylene glycol in water (35% v/v protects down to $-17.6°$ C). $K_{f(H_2O)} = 1.86$.
Osmotic Pressure	$\Pi = iCRT = i\left(\frac{n_2}{V}\right)RT$	Best method for polymers / biomolecules: measured at room temperature; uses molarity; high magnitude.
Van 't Hoff Factor (i)	$i = \frac{M_{\text{normal}}}{M_{\text{obs}}} = \frac{\text{Observed}}{\text{Calculated}}$	Dissociation: $i = 1 + (n-1)\alpha \quad (i > 1)$. Association: $i = 1 - \left(1 - \frac{1}{n}\right)\beta \ (i < 1)$.

TOP 7 HIGH-FREQUENCY CUET TRAPS & PITFALLS

1. **Temperature Dependence of Concentration:** Molarity (M) and Normality (N) change with temperature because liquid volume expands with heating ($d\frac{V}{d}T > 0$). In contrast, Molality (m), Mole Fraction (x), and Mass Percentage (% w/w) are based purely on mass and are **strictly temperature-independent**.
2. **Henry's Constant (K_H) vs Solubility Paradox:** Henry's law states $p = K_H x$. Therefore, at constant gas pressure p, a **higher K_H corresponds to lower gas solubility**. Because K_H increases with rising temperature, aquatic species prefer cold water where dissolved oxygen is substantially higher!
3. **Azeotropic Distillation Failure:** Azeotropes cannot be separated into pure components by fractional distillation because the liquid and vapour phases have identical compositions ($y_1 = x_1$). Pure 100% ethanol cannot be obtained by distilling beer/wine; distillation terminates at the azeotropic composition (95.4% ethanol).
4. **Chloroform-Acetone H-Bond Formation:** Mixing chloroform ($CHCl_3$) and acetone (CH_3COCH_3) releases heat ($\Delta H < 0$) and contracts volume ($\Delta V < 0$) because a new, strong intermolecular hydrogen bond ($Cl_3C - H \cdots O = C(CH_3)_2$) forms between dissimilar molecules (negative deviation).
5. **Freezing Point vs Depression Distinction:** For electrolyte solutions, higher i produces a larger **depression** in freezing point (ΔT_f), which means the **actual freezing point ($T_f = T_f^0 - \Delta T_f$) is the LOWEST**. 0.1 M $Al_2(SO_4)_3$ has the lowest freezing point, while 0.1 M glucose has the highest freezing point!
6. **Osmotic Pressure Supremacy for Macromolecules:** Osmotic pressure is universally preferred over boiling elevation or freezing depression for measuring molecular masses of proteins, polymers, and biomolecules because: (a) osmotic pressure is measurable at room temperature (preventing protein denaturation), and (b) molarity changes produce huge, readily readable hydrostatic heads.
7. **Dimerization of Carboxylic Acids in Benzene:** Benzoic acid or acetic acid dissolved in non-polar solvents like benzene undergoes near-complete dimerization via two intermolecular hydrogen bonds ($2C_6H_5COOH \rightarrow (C_6H_5COOH)_2$). The particle count halves ($i \approx 0.5$), doubling observed molar mass!

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. Which of the following mixtures represents an example of a solid solution where a liquid is dissolved in a solid?

(A) Sodium amalgam (mercury dissolved in sodium)
(B) Brass alloy (copper dissolved in zinc)
(C) Camphor in nitrogen gas
(D) Hydrated copper sulphate crystals

Sol. Q1. In sodium amalgam, liquid mercury (Hg) is dispersed homogeneously into solid sodium metal (Na). Thus, mercury is the liquid solute and sodium is the solid solvent, forming a liquid-in-solid solution. Brass is solid-in-solid, and camphor in nitrogen is solid-in-gas.

Answer: (A)

Q2. A solution is prepared by dissolving 4.0 g of $NaOH$ (molar mass = 40 g/mol) in 250 g of pure water. The molality (m) of the resulting solution is:

(A) 0.10 m
(B) 0.40 m
(C) 0.16 m
(D) 0.04 m

Sol. Q2. Moles of solute $NaOH$: $n_2 = \frac{w_2}{M_2} = \frac{4.0}{40} = 0.10$ mol. Mass of solvent $w_1 = 250$ g $= 0.250$ kg. Molality: $m = \frac{n_2}{W_1}(\text{kg}) = \frac{0.10}{0.250} = 0.40$ mol/kg $= 0.40$ m.

Answer: (B)

Q3. Which of the following concentration metrics changes with a change in temperature?

(A) Molality (m)
(B) Mole fraction (x)
(C) Molarity (M)
(D) Mass percentage (% w/w)

Sol. Q3. Molarity is defined as moles of solute per litre of solution. Since volume expands with temperature ($d\frac{V}{d}T > 0$), molarity decreases with rising temperature. Molality, mole fraction, and mass percentage depend purely on masses and are temperature-independent.

Answer: (C)

Q4. According to Henry's law, the solubility of a gas in a liquid is directly proportional to its partial pressure ($p = K_H x$). Which statement about Henry's constant K_H is strictly correct?

(A) Higher K_H value indicates higher solubility of the gas in the liquid
(B) K_H remains constant regardless of the temperature of the solution
(C) Gases with lower K_H values are less soluble at the same partial pressure
(D) Higher K_H value indicates lower solubility; K_H increases with temperature

Sol. Q4. From $x = \frac{p}{K_H}$, at constant pressure, a larger K_H produces a smaller mole fraction x (lower solubility). As temperature increases, gas solubility decreases, meaning K_H increases with rising temperature. Hence aquatic species thrive in cold water.

Answer: (D)

Q5. Scuba divers often encounter decompression sickness ('the bends') when surfacing rapidly. Deep-sea diving air cylinders are diluted with helium gas because helium:

(A) Has exceptionally low solubility in blood even under high pressure
(B) Acts as an oxygen activator at low oceanic temperatures
(C) Forms a toxic complex with blood hemoglobin
(D) Reacts exothermically with blood plasma to maintain body heat

Sol. Q5. High underwater pressure forces nitrogen into blood. Rapid surfacing causes nitrogen bubbles to block capillaries. Helium is added ($11.7\%He + 56.2\%N_2 + 32.1\%O_2$) because it has extremely low blood solubility, avoiding nitrogen narcosis and the bends.

Answer: (A)

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. Two liquids A and B form an ideal binary solution. At 300 K, $p_A^0 = 100$ Torr and $p_B^0 = 200$ Torr. If the liquid phase has mole fraction $x_A = 0.40$, the mole fraction of A in the vapour phase (y_A) is:

(A) 0.40 **(B)** 0.25
(C) 0.60 **(D)** 0.75

Sol. Q6. $p_A = 0.40 \times 100 = 40$ Torr, $p_B = 0.60 \times 200 = 120$ Torr. Total pressure $P = 40 + 120 = 160$ Torr. Mole fraction in vapour: $y_A = \frac{p_A}{P} = \frac{40}{160} = 0.25$.

Answer: (B)

Q7. A mixture of chloroform ($CHCl_3$) and acetone (CH_3COCH_3) exhibits negative deviation from Raoult's law. What thermal and volume changes occur during mixing?

(A) $\Delta H > 0$ and $\Delta V > 0$ (endothermic, volume expansion) **(B)** $\Delta H > 0$ and $\Delta V < 0$ (endothermic, volume contraction)
(C) $\Delta H < 0$ and $\Delta V > 0$ (exothermic, volume expansion) **(D)** $\Delta H < 0$ and $\Delta V < 0$ (exothermic, volume contraction)

Sol. Q7. Forms strong new $Cl_3C - H\cdots O = C(CH_3)_2$ H-bonds, releasing heat ($\Delta H < 0$) and contracting volume ($\Delta V < 0$).

Answer: (D)

Q8. Which of the following binary liquid pairs forms a minimum boiling azeotrope?

(A) Nitric acid and Water (68%HNO_3) **(B)** Chloroform and Acetone
(C) Ethanol and Water (95.4% ethanol by volume) **(D)** Benzene and Toluene

Sol. Q8. Solutions exhibiting large positive deviations from Raoult's law form minimum boiling azeotropes at a specific intermediate composition. Ethanol-water (95.4% ethanol) boils at 351.15 K, below both pure ethanol (351.5 K) and pure water (373.15 K).

Answer: (C)

Q9. The vapour pressure of pure water at 293 K is 17.535 mm Hg. When 25 g of glucose ($C_6H_{12}O_6$, molar mass = 180 g/mol) is dissolved in 450 g of water, the relative lowering of vapour pressure is:

(A) 0.0055 **(B)** 0.055
(C) 0.0125 **(D)** 0.125

Sol. Q9. Moles of glucose $n_2 = \frac{25}{180} = 0.1389$ mol. Moles of water $n_1 = \frac{450}{18} = 25.0$ mol. Relative lowering of vapour pressure: $\frac{p^0 - p}{p^0} \approx \frac{n_2}{n_1} = \frac{0.1389}{25.0} = 0.00555 \approx 0.0055$.

Answer: (A)

Q10. The molal boiling point elevation constant (K_b) of a solvent depends exclusively on:

(A) The nature of the solute dissolved in it **(B)** The molarity and molality of the solute particles
(C) The nature and thermodynamic properties of the solvent **(D)** The atmospheric pressure and ambient room temperature

Sol. Q10. $K_b = \frac{RM_1T_b^2}{1000\Delta_{vap}H}$, which is an intensive thermodynamic constant determined entirely by the molecular weight, normal boiling point, and enthalpy of vaporization of the solvent alone.

Answer: (C)

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. Which of the following 0.10 M aqueous solutions exhibits the LOWEST freezing point?

(A) 0.10 M Glucose ($C_6H_{12}O_6$) **(B)** 0.10 M $NaCl$
(C) 0.10 M $Al_2(SO_4)_3$ **(D)** 0.10 M K_2SO_4

Sol. Q11. $\Delta T_f = iK_f m$. Since $T_f = T_f^0 - \Delta T_f$, the solution with the HIGHEST i suffers the largest depression, giving the LOWEST freezing point: $Al_2(SO_4)_3$ dissociates into 5 ions ($2Al^{3+} + 3SO_4^{2-}$), so $i = 5$. Thus it has the lowest freezing point.

Answer: (C)

Q12. Why is osmotic pressure (Π) universally preferred over freezing point depression (ΔT_f) or boiling elevation (ΔT_b) for determining the molar masses of proteins and synthetic polymers?

(A) Proteins are insoluble in cold water at freezing temperatures

(B) Measurements can be conducted at room temperature where biomolecules remain stable, and produce large measurable values even for dilute solutions

(C) The van 't Hoff factor i is always exactly zero for macromolecular solutions

(D) Osmotic pressure depends on the chemical nature rather than particle count

Sol. Q12. Proteins denature and degrade at high boiling temperatures. Because polymers have high molar masses, their molality is miniscule, giving imperceptible ΔT_f values ($\approx 10^{-3}$ K). In contrast, osmotic pressure gives measurable heads ($\Pi = CRT$) at room temperature.

Answer: (B)

Q13. What happens to human red blood cells (RBCs) when suspended in an aqueous solution containing 1.5%(w/v)$NaCl$? (Normal saline isotonic with blood plasma is 0.9%(w/v)).

(A) Water flows out of the erythrocytes via exosmosis, causing cells to shrink and crenate

(B) Water enters the erythrocytes via endosmosis, causing cells to swell and burst

(C) No net movement of water occurs because the membrane is impermeable to water

(D) Sodium ions diffuse into the cell until ionic equilibrium is established

Sol. Q13. 1.5%$NaCl$ is hypertonic relative to intracellular plasma (0.9%$NaCl$). Water diffuses out of the cell across the semipermeable cell membrane by exosmosis into the more concentrated external solution, causing the red blood cells to shrink and shrivel (crenation).

Answer: (A)

Q14. A 0.10 m aqueous solution of potassium ferricyanide ($K_3[Fe(CN)_6]$) has a freezing point of $-0.651°$ C. Given $K_f = 1.86$ K kg mol^{-1}, the experimental van 't Hoff factor i and percentage dissociation α are:

(A) $i = 3.0, \alpha = 66.7\%$ **(B)** $i = 3.5, \alpha = 83.3\%$
(C) $i = 4.0, \alpha = 100\%$ **(D)** $i = 2.5, \alpha = 50.0\%$

Sol. Q14. $\Delta T_f = 0.651$ K. $\Delta T_f = iK_f m \Rightarrow 0.651 = i \times 1.86 \times 0.10 \Rightarrow i = \frac{0.651}{0.186} = 3.50$. For $K_3[Fe(CN)_6]$, $n = 3 + 1 = 4$ ions. $\alpha = \frac{i-1}{n-1} = \frac{3.50-1}{4-1} = \frac{2.50}{3} = 0.833 = 83.3\%$.

Answer: (B)

Q15. When benzoic acid (C_6H_5COOH, molar mass = 122 g/mol) is dissolved in benzene, it undergoes dimerization. If the degree of association is $\beta = 0.80$, the observed experimental molar mass is approximately:

(A) 122 g/mol **(B)** 61 g/mol
(C) 150 g/mol **(D)** 203 g/mol

Sol. Q15. Dimerization: $n = 2$. $i = 1 - \left(1 - \frac{1}{n}\right)\beta = 1 - \left(1 - \frac{1}{2}\right)(0.80) = 1 - 0.40 = 0.60$. Van 't Hoff factor: $i = \frac{M_{\text{normal}}}{M_{\text{obs}}} \Rightarrow M_{\text{obs}} = \frac{M_{\text{normal}}}{i} = \frac{122}{0.60} \approx 203.3$ g/mol.

Answer: (D)

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